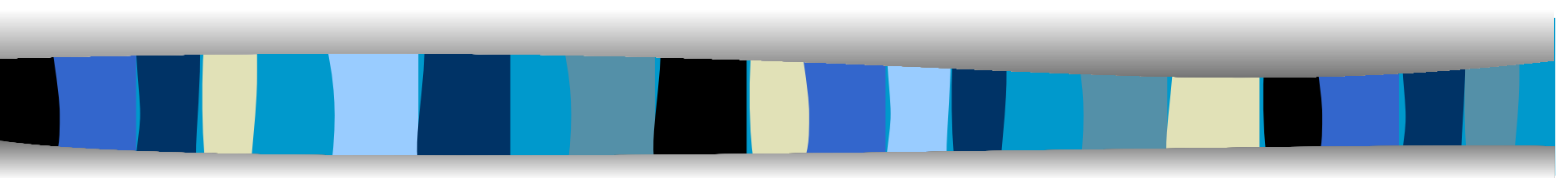


Abstract Algebra



Modulo arithmetic, Group, Ring, Field

Set of Integers

- The set of integers, denoted by $\mathbb{Z}$, contains all integral numbers (with no fraction) from negative infinity to positive infinity
- $\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$

Add:	$5 + 9 = 14$	$(-5) + 9 = 4$	$5 + (-9) = -4$	$(-5) + (-9) = -14$
Subtract:	$5 - 9 = -4$	$(-5) - 9 = -14$	$5 - (-9) = 14$	$(-5) - (-9) = +4$
Multiply:	$5 \times 9 = 45$	$(-5) \times 9 = -45$	$5 \times (-9) = -45$	$(-5) \times (-9) = 45$

Integer Division

- If we divide a by n , we can get q and r .
- The relationship between these four integers can be shown as

- $a = q \cdot n + r$

$$\begin{array}{r}
 n \longrightarrow 11 \quad \left| \begin{array}{r}
 23 \longleftarrow q \\
 \hline
 255 \longleftarrow a \\
 22 \\
 \hline
 35 \\
 33 \\
 \hline
 2 \longleftarrow r
 \end{array} \right.
 \end{array}$$

- 2) -6 (3
- 6
- -12

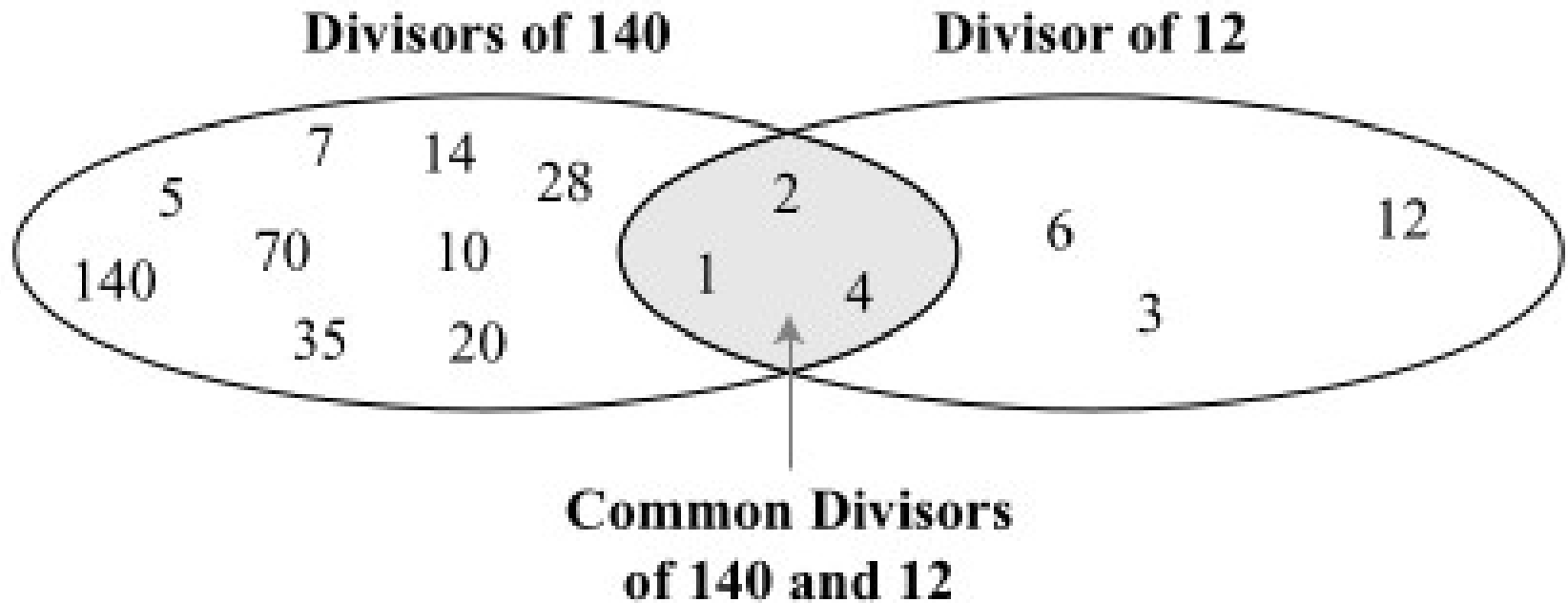
Divisibility

- In cryptography, we take
- $n > 0$
- $r \geq 0$
- a is divisible by n if $r = 0$.
- $a = nq$
- We say n divides a .
- Written as $n|a$
- $4|8$ $12|24$

Greatest Common Divisor

- The greatest common divisor of a and b is the largest integer that divides both a and b .
- We denote with $\gcd(a,b)$
- $\gcd(0, 0) = 0$
- Two positive integers may have many common divisors, but only one greatest common divisor.
- For example, the common divisors of 12 and 140 are 1, 2, and 4.
- However, the greatest common divisor is 4.

Greatest Common Divisor



Euclidean Algorithm

- Finding the greatest common divisor (gcd) of two positive integers by listing all common divisors is not practical when the two integers are large.
- Fortunately, more than 2000 years ago a mathematician named Euclid developed an algorithm that can find the greatest common divisor of two positive integers.
- The Euclidean algorithm is based on the following two facts
- **Fact 1:** $\text{gcd}(a, 0) = a$
- **Fact 2:** $\text{gcd}(a, b) = \text{gcd}(b, r)$, where r is the remainder of dividing a by b

Euclidean Algorithm

- $\gcd(36, 10) = \gcd(10, 6) = \gcd(6, 4) = \gcd(4, 2) = \gcd(2, 0) = 2$

q	r_1	r_2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	20	0	

Euclidean Algorithm

q	r_1	r_2	r
0	25	60	25
2	60	25	10
2	25	10	5
2	10	5	0
	5	0	

Extended Euclidean Algorithm

- The greatest common divisor between two integers a and b can be expressed as the linear combination of two integers s and t .
- i. e. $\gcd(a,b) = sa + tb$
- These two integers can be found using Extended Euclidean Algorithm.
- This algorithm not only finds gcd but also two integers s and t .
- It is similar to Euclidean algorithm but it uses additional variables $s1, s2, s, t1, t2$ and t .
- $s1, s2$ are initialized to 1, 0

Extended Euclidean Algorithm

q	a	n	r	s1	s2	$s=s1-s2*q$	t1	t2	$t=t1-t2*q$
1	252	198	54	1	0	1	0	1	-1
3	198	54	36	0	1	-3	1	-1	4
1	54	36	18	1	-3	4	-1	4	-5
2	36	18	0	-3	4	-11	4	-5	14
	18	0		4	-11	4	-5	14	-5

Linear Diophantine Equations

- One of the application of extended euclidean algorithm is to find the solutions to diophantine equations.
- Diophantine equations of two variables is $ax + by = c$.
- We need to find integer values for x and y that satisfy the equation.
- This type of equation has either no solution or an infinite number of solutions. Let $d = \gcd(a, b)$.
- If $d \mid c$, then we have an infinite number of solutions.
- One of them is called the particular solution.
- Other solutions are generic.

Linear Diophantine Equations

- **Steps for particular solution**
- Reduce the equation $ax + by = c$ to $a_1x + b_1y = c_1$ by dividing both sides by d .
- Here $a_1 = a/d$, $b_1 = b/d$, $c_1 = c/d$
- Get the values of s and t using a and b such that $a_1s + b_1t = 1$ using the extended Euclidean algorithm.
- The particular solution can be found:
- $x_0 = c_1s$ and $y_0 = c_1t$
- General solutions: $x = x_0 + k b_1$ and
- $y = y_0 - k a_1$ where k is an integer

Linear Diophantine Equations

- Find the particular and general solutions to the equation $21x + 14y = 35$.
- We have $d = \gcd(21, 14) = 7$.
- Since $7 \mid 35$, the equation has an infinite number of solutions.
- We can divide both sides by 7 to find the equation $3x + 2y = 5$.
- We get $a_1=3$, $b_1=2$ and $c_1=5$

Linear Diophantine Equations

- Now, we use extended Euclidean algorithm on $a_1(3)$ and $b_1(2)$ to find s and t such that $a_1s + b_1t = 1$.

q	a	n	r	s1	s2	s=s1-s2*q	t1	t2	t=t1-t2*q
1	3	2	1	1	0	1	0	1	-1
2	2	1	0	0	1	-2	1	-1	3
	1	0		1	-2	1	-1	3	-1

- From the table, we can see that $s=1$ and $t=-1$
- Particular Solution is $c/d=35/7=5$
- $x_0 = (c/d)s=5 \times 1 = 5$ and $y_0 = (c/d)t=5 \times (-1) = -5$

Linear Diophantine Equations

- Now, we use extended Euclidean algorithm on $a_1(3)$ and $b_1(2)$ to find s and t such that $a_1s + b_1t = 1$. We have $s = 1$ and $t = -1$.
- Particular Solution is $c/d=35/7=5$
- $x_0 = (c/d)s=5 \times 1 = 5$ and $y_0 = (c/d)t=5 \times (-1) = -5$
- $(5, -5), (7, -8), (9, -11) \dots$ are some of the solutions
- $x=x_0+kb_1 = 5+1*2=7$
- $y=y_0+ka_1=-5-3=-8$

Modulo Arithmetic

- If a is an integer and n is a positive integer, we define $a \bmod n$ to be the remainder when a is divided by n .
- For integer a , we can write the equation
- $a = q * n + r$
- This equation needs two input, a and n .
- It has two output q and r .
- In modulo arithmetic, the important output is the remainder r .
- We are not interest in quotient q .

Modulo Arithmetic

- In modulo arithmetic, we use modulo operator mod to find the remainder.
- $r = a \bmod n$
- Which means r is residue or remainder when a is divided by n .

$$a = qn + r \quad 0 \leq r < n; q = \lfloor a/n \rfloor$$

$$a = \lfloor a/n \rfloor \times n + (a \bmod n)$$

$$11 \bmod 7 = 4; \quad -11 \bmod 7 = 3$$

Modulo Arithmetic

- $27 \bmod 5 = 2$ or -3
- $36 \bmod 12 = 0$ or -12
- $-18 \bmod 14$
- $q = \text{floor}(-18/14) = -1$
- $a = qn + r$ or $-18 = -1 \cdot 14 + r$ or $r = -4$ or 12
- $-7 \bmod 10$
- $q = \text{floor}(-7/10) = 0$
- $-7 = 0 \cdot 10 + r$ or $r = -7$ or 3

Set of Residue

- The result of modulo operation with modulus n is always between 0 and $n-1$.
- $a \bmod n$ is always a nonnegative integer less than n .
- This set of nonnegative integers less than n is called set of residues
- It is represented by Z_n
- $Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$
- $Z_n = \{0, 1, 2, \dots, n-1\}$

Class of Residue

- For each member of set of residue, we can have class of residue.
- For e.g. if b is member of set of residue, its class of residue is represented with $[b]$
- The class of residue $[b]$ is the set of all integers when divided by n gives b as remainder.
- For e.g. $n=5$
- Set of residue is $\{0, 1, 2, 3, 4\}$
- Residue classes are $[0], [1], [2], [3], [4]$ which are given below

Class of Residue

$$[0] = \{\dots, -15, -10, -5, 0, 5, 10, 15, \dots\}$$

$$[1] = \{\dots, -14, -9, -4, 1, 6, 11, 16, \dots\}$$

$$[2] = \{\dots, -13, -8, -3, 2, 7, 12, 17, \dots\}$$

$$[3] = \{\dots, -12, -7, -2, 3, 8, 13, 18, \dots\}$$

$$[4] = \{\dots, -11, -6, -1, 4, 9, 14, 19, \dots\}$$

Operation in set of Residue

Perform the following operations (the inputs come from $\mathbf{Z}_n$):

- a. Add 7 to 14 in $\mathbf{Z}_{15}$.
- b. Subtract 11 from 7 in $\mathbf{Z}_{13}$.
- c. Multiply 11 by 7 in $\mathbf{Z}_{20}$.

Solution

The following shows the two steps involved in each case:

$$(14 + 7) \bmod 15 \rightarrow (21) \bmod 15 = 6$$

$$(7 - 11) \bmod 13 \rightarrow (-4) \bmod 13 = 9$$

$$(7 \times 11) \bmod 20 \rightarrow (77) \bmod 20 = 17$$

Modulus property

First Property: $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$

Second Property: $(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$

Third Property: $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

$10^n \bmod x = (10 \bmod x)^n \bmod x$ Applying the third property n times.

Modulus property

$$11 \bmod 8 = 3; 15 \bmod 8 = 7$$

$$[(11 \bmod 8) + (15 \bmod 8)] \bmod 8 = 10 \bmod 8 = 2$$

$$(11 + 15) \bmod 8 = 26 \bmod 8 = 2$$

$$[(11 \bmod 8) - (15 \bmod 8)] \bmod 8 = -4 \bmod 8 = 4$$

$$(11 - 15) \bmod 8 = -4 \bmod 8 = 4$$

$$[(11 \bmod 8) \times (15 \bmod 8)] \bmod 8 = 21 \bmod 8 = 5$$

$$(11 \times 15) \bmod 8 = 165 \bmod 8 = 5$$

Modulus property

Property	Expression
Commutative Laws	$(w + x) \bmod n = (x + w) \bmod n$ $(w \times x) \bmod n = (x \times w) \bmod n$
Associative Laws	$[(w + x) + y] \bmod n = [w + (x + y)] \bmod n$ $[(w \times x) \times y] \bmod n = [w \times (x \times y)] \bmod n$
Distributive Law	$[w \times (x + y)] \bmod n = [(w \times x) + (w \times y)] \bmod n$
Identities	$(0 + w) \bmod n = w \bmod n$ $(1 \times w) \bmod n = w \bmod n$
Additive Inverse ($-w$)	For each $w \in Z_n$, there exists a z such that $w + z \equiv 0 \bmod n$

Calculating exponent modulo n

- $2^3 \bmod 5$
- $=3$
- $2^3 \bmod 5$
- $-3^3 \bmod 5$
- $-27 \bmod 5$
- $-2 \bmod 5$
- 3

Calculating exponent modulo n

- $9^{38} \bmod 5$
- $= (9 \bmod 5)^{38} \bmod 5$
- $= 4^{38} \bmod 5$
- $= -1^{38} \bmod 5$
- $= 1 \bmod 5$
- $= 1$

Calculating exponent modulo n

- $125^{107} \bmod 187$
- Binary of 107 is 1101011 that is 64, 32, 8, 2, 1
- $=125^{64} 125^{32} 125^8 125^2 125 \bmod 187$
- Lets calculate $125^2 \bmod 187 = 104$
- $125^4 \bmod 187 = 104^2 \bmod 187 = 157$
- $125^8 \bmod 187 = (125^4)^2 \bmod 187$
- $= 157^2 \bmod 187 = 152$
- $125^{16} \bmod 187 = (125^8)^2 \bmod 187$
- $= 152^2 \bmod 187 = 103$

Calculating exponent modulo n

- $125^{32} \bmod 187 = (125^{16})^2 \bmod 187$
- $= 103^2 \bmod 187 = 137$
- $125^{64} \bmod 187 = (125^{32})^2 \bmod 187$
- $= 137^2 \bmod 187 = 69$
- Finally
- $125^{64} 125^{32} 125^8 125^2 125 \bmod 187$
- $69 \times 137 \times 152 \times 104 \times 125 \bmod 187$
- $= [(69 \times 137 \bmod 187) (152 \times 104 \bmod 187)] 125 \bmod$

Calculating exponent modulo n

- $=[(69 \times 137 \bmod 187) (152 \times 104 \bmod 187)]125 \bmod 187$
- $=103 \times 100 \times 125 \bmod 187$
- $=15 \times 125 \bmod 187$
- $=5$

Calculating exponent modulo n

- $a^m \bmod n$
- is calculated as given below
- $z = 1$
- Calculate binary form b of m
- for $i = k$ down to 0 do
 - $z = z^2 \bmod n$
 - if $b_i = 1$ then $z = z \times a \bmod n$
- So final answer is z .

Calculating exponent modulo n

- $125^{107} \bmod 187$, here, $a=125$, $m=107$, $n=187$

i	bit of m	$z=z^2 \bmod n$	$z=z*a \bmod n$
6	1	$1^2 \bmod 187=1$	$1*125 \bmod 187=125$
5	1	$125^2 \bmod 187=104$	$104*125 \bmod 187=97$
4	0	$97^2 \bmod 187=59$	59
3	1	$59^2 \bmod 187=115$	$115*125 \bmod 187=163$
2	0	$163^2 \bmod 187=15$	15
1	1	$15^2 \bmod 187=38$	$38*125 \bmod 187=75$
0	1	$75^2 \bmod 187=15$	$15*125 \bmod 187=5$

Calculating exponent modulo n

- $9^{38} \bmod 5$, $n=5$, $m=38$, $a=9$

i	b_i of m	$z=z^2 \bmod n$	$z=z*a \bmod n$
5	1	$1^2 \bmod 5=1$	$1*9 \bmod 5=4$
4	0	$4^2 \bmod 5=1$	1
3	0	$1^2 \bmod 5=1$	1
2	1	$1^2 \bmod 5=1$	$1 * 9 \bmod 5=4$
1	1	$4^2 \bmod 5=1$	$1 * 9 \bmod 5=4$
0	0	$4^2 \bmod 5=1$	1

Calculating exponent modulo n

- $9726^{3533} \bmod 11413 = 5761$

i	b_i	z
11	1	$1^2 \times 9726 = 9726$
10	1	$9726^2 \times 9726 = 2659$
9	0	$2659^2 = 5634$
8	1	$5634^2 \times 9726 = 9167$
7	1	$9167^2 \times 9726 = 4958$
6	1	$4958^2 \times 9726 = 7783$
5	0	$7783^2 = 6298$
4	0	$6298^2 = 4629$
3	1	$4629^2 \times 9726 = 10185$
2	1	$10185^2 \times 9726 = 105$
1	0	$105^2 = 11025$
0	1	$11025^2 \times 9726 = 5761$

Congruence

- Two integers a and b are said to be **congruent modulo n** ,
- if $(a \bmod n) = (b \bmod n)$.
- Also, if $n|(a-b)$ and $n|(b-a)$, then a and b are congruent modulo n .
- This is written as $a \equiv b \pmod{n}$
- $73 \equiv 4 \pmod{23}$;
- $21 \equiv 31 \pmod{10}$
- $21 \equiv -9 \pmod{10}$

Properties of Congruence

1. $a \equiv b \pmod{n}$ if $n \mid (a - b)$.
2. $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n}$.
3. $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ imply $a \equiv c \pmod{n}$.

$23 \equiv 8 \pmod{5}$	because	$23 - 8 = 15 = 5 \times 3$
$-11 \equiv 5 \pmod{8}$	because	$-11 - 5 = -16 = 8 \times (-2)$
$81 \equiv 0 \pmod{27}$	because	$81 - 0 = 81 = 27 \times 3$

Additive Inverse

- If result of addition of two integers a and b modulo n is additive identity(0) then a and b are called additive inverses of each other.
- $a+b \equiv 0 \pmod{n}$
- $0+0 \equiv 0 \pmod{10}$
- $1+9 \equiv 0 \pmod{10}$
- $2+8 \equiv 0 \pmod{10}$

Multiplicative Inverse

- Two integers a and b are multiplicative inverses of each other modulo n if
- $ab \equiv 1 \pmod{n}$
- a has multiplicative inverse modulo n only if a and n are co-prime or relatively prime.
- 8 has no multiplicative inverse modulo 10
- Since $\gcd(8,10) = 2$

Multiplicative Inverse

- Find multiplicative inverse of 3 modulo 10.
- $3 * 1 \bmod 10 = 3$
- $3 * 2 \bmod 10 = 6$
- $3 * 3 \bmod 10 = 9$
- $3 * 4 \bmod 10 = 2$
- $3 * 5 \bmod 10 = 5$
- $3 * 6 \bmod 10 = 8$
- $3 * 7 \bmod 10 = 1$
- So, 7 is multiplicative inverse of 3 modulo 10.

Multiplicative Inverse

- We can find multiplicative using extended euclidean algorithm.
- It will calculate $\gcd(n,b)$ to show that multiplicative inverse for $b \bmod n$ exists.
- For this we will use similar table as done in extended euclidean algorithm.
- we are only interested in t not in s .
- The last value of $t1$ will be multiplicative inverse of b .
- We use a as n and n as b .

Multiplicative Inverse

- Find multiplicative inverse of 11 mod 26

q	n	b	r	t1	t2	t=t1-t2*q
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
	1	0		-7	26	-7

- As we can see that $\gcd(26, 11) = 1$
- So, multiplicative inverse of 11 mod 26 exists.
- Its $-7 \bmod 26 = 19$
- We see that $19 * 11 \bmod 26 = 209 \bmod 26 = 1$

Multiplicative Inverse

- Find multiplicative inverse of 23 mod 100

q	n	b	r	t1	t2	t=t1-t2*q
4	100	23	8	0	1	-4
2	23	8	7	1	-4	9
1	8	7	1	-4	9	-13
7	7	1	0	9	-13	100
	1	0		-13	100	-13

- As we can see that $\gcd(100, 23) = 1$
- So, multiplicative inverse of 23 mod 100 exists.
- Its $-13 \bmod 100 = 87$
- We see that $87 * 23 \bmod 100 = 2001 \bmod 100 = 1$

Multiplicative Inverse

Example 2.27

Find the inverse of 12 in $\mathbf{Z}_{26}$.

Solution

We use a table similar to the one we used before, with $r_1 = 26$ and $r_2 = 12$.

q	r_1	r_2	r	t_1	t_2	t
2	26	12	2	0	1	-2
6	12	2	0	1	-2	13
	2	0		-2	13	

The $\gcd(26, 12) = 2 \neq 1$, which means there is no multiplicative inverse for 12 in $\mathbf{Z}_{26}$.

Addition and Multiplication Table

Addition and multiplication tables for $\mathbf{Z}_{10}$

	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	0
2	2	3	4	5	6	7	8	9	0	1
3	3	4	5	6	7	8	9	0	1	2
4	4	5	6	7	8	9	0	1	2	3
5	5	6	7	8	9	0	1	2	3	4
6	6	7	8	9	0	1	2	3	4	5
7	7	8	9	0	1	2	3	4	5	6
8	8	9	0	1	2	3	4	5	6	7
9	9	0	1	2	3	4	5	6	7	8

Addition Table in $\mathbf{Z}_{10}$

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	1	8	0	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	1

Multiplication Table in $\mathbf{Z}_{10}$

Z_n and Z_n^*

- Z_n^* is subset of Z_n
- It includes the elements that have multiplicative inverse in Z_n

Some Z_n and Z_n^ sets*

$$Z_6 = \{0, 1, 2, 3, 4, 5\}$$

$$Z_6^* = \{1, 5\}$$

$$Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$$

$$Z_7^* = \{1, 2, 3, 4, 5, 6\}$$

$$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$Z_{10}^* = \{1, 3, 7, 9\}$$

Z_p and Z_{p^*}

- Z_p is same to Z_n except p is prime.
- Z_{p^*} is subset of Z_p which includes elements in Z_p that has both additive and multiplicative inverse in Z_p .
- It means it includes all the elements from 1 to $p-1$
- 0 has no multiplicative inverse.
- $Z_{13} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
- $Z_{13^*} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

Linear Congruence

- One of the application of congruence is to solve the equations.
- Let us consider the equation of the form
- $ax \equiv b \pmod{n}$.
- An equation of this type might have no solution or a limited number of solutions.
- Let $d = \gcd(a, n)$.
- If d does not divide b , there is no solution.
- If $d \mid b$, there are d solutions.

Linear Congruence

- The general solutions are $x = x_0 + k(n / d)$ for $k = 0, 1, \dots, (d - 1)$.
- Solve the equation $14x \equiv 12 \pmod{18}$.
- $d = \gcd(14 \text{ and } 18) = 2$. Since 2 divides 12, we have exactly two solutions
- $14x \equiv 12 \pmod{18} \rightarrow 7x \equiv 6 \pmod{9}$
- $x \equiv 6(7^{-1}) \pmod{9}$
- $x_0 = (6 \times 7^{-1}) \pmod{9} = (6 \times 4) \pmod{9} = 6$
- $x_1 = x_0 + 1 \times (18/2) = 15$

Figure 2.27 *Set of linear equations*

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & \equiv & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & \equiv & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{n1}x_1 & + & a_{n2}x_2 & + & \dots & + & a_{nn}x_n & \equiv & b_n \end{array}$$

a. Equations

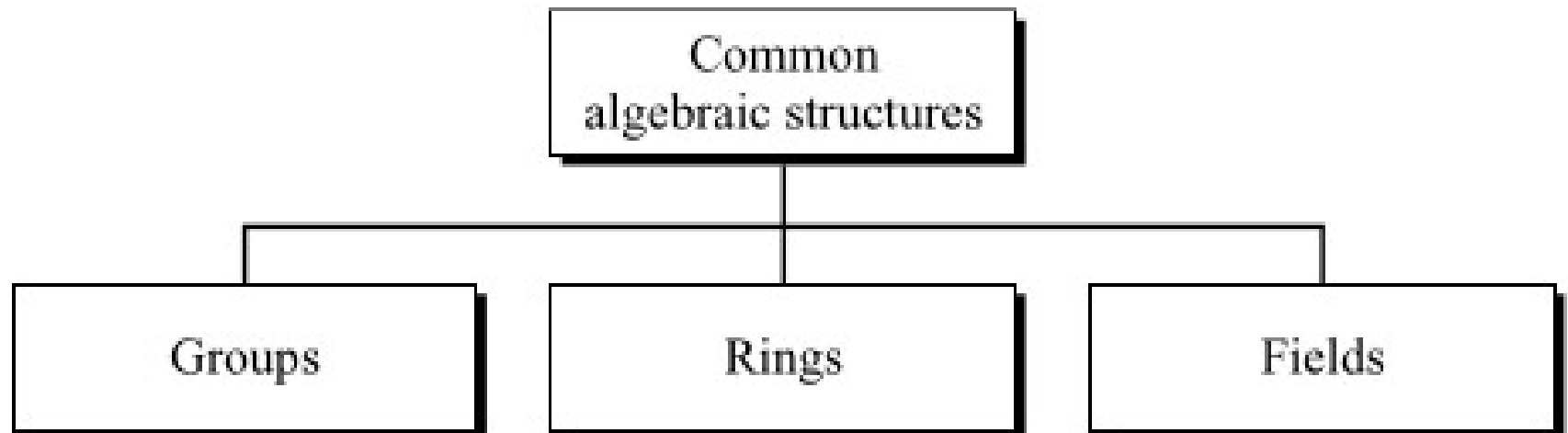
$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \equiv \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$
$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \equiv \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}^{-1} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

b. Interpretation

c. Solution

Algebraic Structures

- The non empty set together with one or more binary operations on the elements of the set that satisfies certain properties are called algebraic structures.
- $\langle G, \$ \rangle$ is a structure where G is non empty set and $\$$ is operation defined one elements of G .



Group

- An algebraic structure $\langle G, \bullet \rangle$ is said to be group if the binary operation $\bullet$ satisfies following properties for all elements a, b, c of G .
- **Associative law** $(a \bullet b) \bullet c = a \bullet (b \bullet c)$
- **Identity element e exists** $a \bullet e = e \bullet a = a$
- For each element **a in G , inverse element a^{-1} exists.** $a \bullet a^{-1} = e$
- **Closure law** $c = a \bullet b$
- Here a, b, c, e are elements of the group G .

Group

- Group can be also defined as the set of elements that is closed with respect to four different operation.
- Closed means the result of the operation is also in the set
- Closed property is related with closure law.

Group

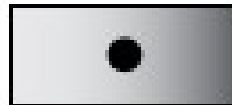
Properties

1. Closure
2. Associativity
3. Commutativity (See note)
4. Existence of identity
5. Existence of inverse

*Note:
The third property needs
to be satisfied only for a
commutative group.*

$\{a, b, c, \dots\}$

Set



Operation

Group

Group

- The set of residue integers with the addition operator, $G = \langle \mathbb{Z}_8, + \rangle$, is a group
- **Closure is satisfied.** The result of multiplying two integers in $\mathbb{Z}_8$ is another integer in $\mathbb{Z}_8$.
- **Associativity is satisfied.**
- $4 + (3 + 2) = (4 + 3) + 2$.
- **Identify element is 0.**
- $3 + 0 = 0 + 3 = 3$.
- **Inverse element of 4 is 4**
- $4 + 4 \bmod 8 = 0$ (inverse of 4 is 4)

Exponentiation of a element in group

- Let us consider a group $\langle G, \bullet \rangle$
- For any element g in G , the power of exponent of g is defined as below.
- $g^0 = e$
- $g^1 = g$
- $g^{-1} =$ inverse of g under $\bullet$
- For any integer $k \geq 2$, power is repetitive application of group operation as shown below
- $g^k = g \bullet g^{k-1}$

Exponentiation of a element in group

- For any integer $k < -1$,
- $g^{-k} = g^{-1} \cdot g^{-(k-1)}$
- $g^m \cdot g^n = g^{m+n}$
- $(g^m)^n = g^{mn}$

Exponentiation of a element in group

- Let us consider a group $G = \langle \mathbb{Z}_6, + \rangle$
- So, elements of $G = \{0, 1, 2, 3, 4, 5\}$
- Let us find the 6 power of element 2
- $2^6 = (2+2+2+2+2+2) \bmod 6 = 2$
- $3^7 = (3+3+3+3+3+3+3) \bmod 6 = 3$
- $3^0 = 0$ (identity element)

Abelian Group

- An abelian group, is a group $\langle G, \bullet \rangle$ in which elements satisfies the commutativity with respect to binary operation $\bullet$.
- Let a and b be elements of the the group.
- If $a \bullet b = b \bullet a$ then G is a abelian group.
- It is also known as commutative group.

Abelian Group

- The group $G = \langle \mathbb{Z}_8, + \rangle$ is an Abelian
- $4+5=5+4$

Abelian Group

- Show that a group $G = \langle \mathbb{Z}_n^*, \times \rangle$ is an Abelian group

Finite Group

- A group is called a **finite group** if the set has a finite number of elements; otherwise, it is an **infinite group**.
- The **order** of the group is equal to the number of elements in the group.
- Some of infinite groups are.
 - the group of real numbers (with addition),
 - the group of integers (with addition),
 - the group of rational numbers (with addition)
- The **order** of infinite group is also infinite.

Subgroup

- A subset H of a group G is a subgroup of G if H itself is a group with respect to the operation on G .
- In other words, if $G = \langle S, \bullet \rangle$ is a group, $H = \langle T, \bullet \rangle$ is a group under the same operation
- Where, T is a nonempty subset of S
- If a and b are members of both groups, then $c = a \bullet b$ is also a member of both groups.
- Both group share the same identity element.

Subgroup

- If a is a member of both groups, the inverse of a is also a member of both groups.
- The group made of the identity element of G , $H = \langle \{e\}, \bullet \rangle$, is a subgroup of G .
- Each group is a subgroup of itself.

Subgroup

- Check if the group $H = \langle \mathbb{Z}_{10}, + \rangle$ a subgroup of the group $G = \langle \mathbb{Z}_{12}, + \rangle$?
- H is not subgroup of G because H has addition operation modulo 10
- Where as G has addition operation modulo 12.
- So, operation is different.

Subgroup

- Let us consider a group, $G = \langle \mathbb{Z}_4, + \rangle$
- One of its subgroups, $H = \langle \{0, 2\}, + \pmod{4} \rangle$
- $\{0, 2\}$ is subset of $\mathbb{Z}_4$.
- Identity element is 0 which is contained in both groups.

Cyclic group

- A group $\langle G, \bullet \rangle$ is a cyclic group if there exists a element g in G such that every element in G can be generated with different powers of g .
- Powers of g means repetitive application of group operation $\bullet$ on g .
- g is said to be generator.
- A cyclic group could have multiple generators.

Cyclic group

- Let us consider a group $G = \langle \mathbb{Z}_6, + \rangle$
- Here group operation is addition modulo 6.
- $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$
- $0^0 = 0$
- $0^1 = 0$
- $0^2 = (0+0) \bmod 6 = 0$
- So, 0 can't be a generator.
- Lets try 1 as generator

Cyclic group

- Let us consider a group $G = \langle \mathbb{Z}_6, + \rangle$
- Here group operation is addition modulo 6.
- $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$
- Lets try 1 as generator
- $1^1 = 1$
- $1^2 = (1+1) \bmod 6 = 2$
- $1^3 = (1+1+1) \bmod 6 = 3$
- $1^4 = (1+1+1+1) \bmod 6 = 4$
- $1^5 = (1+1+1+1+1) \bmod 6 = 5$
- $1^6 = (1+1+1+1+1+1) \bmod 6 = 0$

Cyclic subgroup

- The subgroup of a cyclic group is cyclic.
- Let us consider a cyclic group $G = \langle \mathbb{Z}_6, + \rangle$
- Here group operation is addition modulo 6.
- Its subgroup $\langle \{0\}, + \rangle$, $\langle \{0, 2, 4\}, + \rangle$ $\langle \{0, 3\}, + \rangle$
- Lets consider $\langle \{0, 2, 4\}, + \rangle$
- Let us use 2 as generator
- $2^1 = 2$
- $2^2 = (2 + 2) \bmod 6 = 4$
- $2^3 = (2 + 2 + 2) \bmod 6 = 0$
- So, its a cyclic group with generator 2.

Order of element of a group

- The order of an element a in a group, $\text{ord}(a)$, is the smallest positive integer n such that $a^n = e$.
- The definition can be paraphrased: the order of an element is the order of the cyclic group it generates.
- $G = \langle \mathbb{Z}_6, + \rangle$
- $\text{Ord}(0) = 0^1 = 0$ (identity element)
- $\text{Ord}(1) = 1^1 = 1$ (not identity)
- $\text{Ord}(1) = 1^2 = (1+1) \bmod 6 = 2$ (not identity)
- $\text{Ord}(1) = 1^6 = (1+1+1+1+1+1) \bmod 6 = 0$ (identity)

Ring

- A set R with two binary operations addition (+) and multiplication ($\times$) denoted as $\langle R, +, \times \rangle$ is a ring if following hold
- $\langle R, + \rangle$ is a abelian group.
- For $\times$ operation, it must satisfy **closure**, **associativity**.
- **Closure**: For a, b of R , $a \times b$ must be element of R
- **Associativity**: for elements a, b, c of R ,
- $(a \times b) \times c = a \times (b \times c)$.
- Second operator can be distributed over first
- $a \times (b + c) = (a \times b) + (a \times c)$

Commutative ring

{a,b,.....}	
Operation +	Operation x
1. Closure 2. Associativity 3. Commutativity 4. Existence of Identity element 5. Existence of inverse	1. Closure 2. Associativity
Distribution x over +	

Commutative Ring

- A ring $\langle R, +, \times \rangle$ is commutative ring if multiplication ($\times$) operation satisfies additional commutative property.

Commutative ring

{a,b,.....}	
Operation +	Operation x
1. Closure 2. Associativity 3. Commutativity 4. Existence of Identity element 5. Existence of inverse	1. Closure 2. Associativity 3. Commutativity
Distribution x over +	

Field

- A field, denoted by $F = \langle \{...\}, +, \times \rangle$ is a commutative ring in which the second operation satisfies all five properties defined for the first operation except that the identity of the addition (+) operation has no inverse with respect to multiplication ($\times$).
- $\langle \mathbb{R}, +, \times \rangle$ is a field.
- $\langle \mathbb{Q}, +, \times \rangle$ is a field.
- $\langle \mathbb{Z}_7, +, \times \rangle$ is a field

Field

{a,b,.....}	
Operation +	Operation x
1. Closure 2. Associativity 3. Commutativity 4. Existence of identity element 5. Existence of inverse	1. Closure 2. Associativity 3. Commutativity 4. Existence of identity element 5. Existence of inverse
Distribution x over +	

Finite Field

- A finite field has a finite number of elements,
- They are very extensively used in cryptography.
- Let us consider a field $\langle \mathbb{Z}_7, +, \times \rangle$ under the operations multiplication and addition.
- Set is $\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$
- The first operation is addition modulo 7
- Second operation is multiplication modulo 7.

Galois Fields

- The Galois Field is denoted by $GF(p^n)$ or $GF(p)$ where p is prime and $n=1$.
- Mathematician Galois showed that the order of $GF(p)$ is p .
- So, Galois Field is a finite field.
- Its set is $Z_p = \{0, 1, 2, \dots, p-1\}$
- The first operation is addition modulo p .
- The Second Operation is multiplication modulo p .
- Galois Fields $GF(p)$ are also known as prime field
- They are very extensively used in cryptography.

GF(p) Field

GF(2)

$\{0, 1\}$	$+$	$\cdot$
------------	-----	---------

$+$	0	1
0	0	1
1	1	0

Addition

$\cdot$	0	1
0	0	0
1	0	1

Multiplication

a	0	1
$-a$	1	0

a	0	1
a^{-1}	—	1

Inverses

GF(5)

$\{0, 1, 2, 3, 4\}$	$+$	$\times$
---------------------	-----	----------

$+$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Addition

$\times$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Multiplication

Additive inverse

a	0	1	2	3	4
$-a$	0	4	3	2	1

a	0	1	2	3	4
a^{-1}	—	1	3	2	4

Multiplicative inverse

Galois Fields

- Since, set of $GF(p)$ is set of integers, Both multiplication and addition modulo p satisfies closure, associative, commutative, and existence of identity.
- What we need to check is existence of inverse for each element.
- And it is shown by Galois that for prime p , every element from 0 to $p-1$ has inverse element for both operations.
- Let us inspect for field $GF(7)$ using addition and multiplication table.

Finite Field

+	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

x	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	1	3	5
3	0	3	6	2	5	1	4
4	0	4	1	5	2	6	3
5	0	5	3	1	6	4	2
6	0	6	5	4	3	2	1

- The addition modulo 7 table shows that all elements have additive inverse.
- The multiplication modulo 7 table shows that all elements have multiplicative inverse except 0
- So, $\langle \mathbb{Z}_7, +, \times \rangle$ is a field.

Finite Field

- However $\langle \mathbb{Z}_8, +, \times \rangle$ is not a field as all elements of $\mathbb{Z}_8$ do not have multiplicative inverse.

$\times$	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	0	2	4	6
3	0	3	6	1	4	7	2	5
4	0	4	0	4	0	4	0	4
5	0	5	2	7	4	1	6	3
6	0	6	4	2	0	6	4	2
7	0	7	6	5	4	3	2	1

(b) Multiplication modulo 8

Polynomial

- A polynomial of degree $n-1 \geq 0$ is an expression of the form
- $f(x) = a_{n-1}x^{n-1} + \dots + a_1x^1 + a_0x^0$
- a_i is coefficient of term x^i
- This polynomial can represent n bit binary number.
- The power of x define the position of the bit in the n -bit word.
- The coefficients are actual bits.

Polynomial

- We can denote set of coefficients with S .
- A zero-degree polynomial is called a constant polynomial.
- An n th-degree polynomial is said to be a monic polynomial if $a_n = 1$.

Polynomial Arithmetic

- Polynomial arithmetic deals with the addition, subtraction, multiplication of polynomials.
- Polynomial arithmetic can be done in three different ways.
- **Ordinary polynomial arithmetic**
- **Polynomial arithmetic over a field $\langle S, +, \times \rangle$** where S is some set to which coefficients of polynomial belongs.
- **Polynomial modulo arithmetic $\langle S, +, \times \rangle$** where S is set of polynomials.

Ordinary polynomial Arithmetic

- Polynomial arithmetic operations are defined in a natural way as defined in algebra.
- We can perform addition, multiplication, division, and subtraction of polynomials.

Ordinary Polynomial Arithmetic

As an example, let $f(x) = x^3 + x^2 + 2$ and $g(x) = x^2 - x + 1$, where S is the set of integers. Then

$$f(x) + g(x) = x^3 + 2x^2 - x + 3$$

$$f(x) - g(x) = x^3 + x + 1$$

$$f(x) \times g(x) = x^5 + 3x^2 - 2x + 2$$

Ordinary Polynomial Arithmetic

$$\begin{array}{r} x^3 + x^2 \quad + 2 \\ + (x^2 - x + 1) \\ \hline x^3 + 2x^2 - x + 3 \end{array}$$

(a) Addition

$$\begin{array}{r} x^3 + x^2 \quad + 2 \\ - (x^2 - x + 1) \\ \hline x^3 \quad + x + 1 \end{array}$$

(b) Subtraction

$$\begin{array}{r} x^3 + x^2 \quad + 2 \\ \times (x^2 - x + 1) \\ \hline x^3 + x^2 \quad + 2 \\ - x^4 - x^3 \quad - 2x \\ \hline x^5 + x^4 \quad + 2x^2 \\ \hline x^5 \quad + 3x^2 - 2x + 2 \end{array}$$

(c) Multiplication

$$\begin{array}{r} x^2 - x + 1 \overline{) x^3 + x^2 \quad + 2} \\ \underline{x^3 + x^2 \quad + x} \\ 2x^2 - x + 2 \\ \underline{2x^2 - 2x + 2} \\ x \end{array}$$

(d) Division

Polynomial Arithmetic over Ring

- We can perform polynomial arithmetic over a Ring $\langle S, +, \times \rangle$.
- S is the set to which coefficient belongs.
- All the operations addition, subtraction, multiplication and division may not be supported by the ring.
- However, All the operations are supported over Galois Field $GF(P)$

Polynomial Arithmetic GF(p)

- Given two polynomials $f(x) = (x^7 + x^5 + x^4 + x^3 + x + 1)$ and $g(x) = (x^3 + x + 1)$
- Let us perform polynomial arithmetic over a field GF(2)
- compute $f(x) + g(x)$; $f(x) - g(x)$; $f(x) * g(x)$; and $f(x)/g(x)$. Note that $g(x)|f(x)$.
- The GF(2) consists of coefficient $\{0,1\}$
- In this field addition and subtraction is XOR operation
- Multiplication is modulo 2

Polynomial Arithmetic over GF(2)

$$\begin{array}{r} x^7 \quad + x^5 + x^4 + x^3 \quad + x + 1 \\ \quad \quad \quad + (x^3 \quad + x + 1) \\ \hline x^7 \quad + x^5 + x^4 \end{array}$$

(a) Addition

$$\begin{array}{r} x^7 \quad + x^5 + x^4 + x^3 \quad + x + 1 \\ \quad \quad \quad - (x^3 \quad + x + 1) \\ \hline x^7 \quad + x^5 + x^4 \end{array}$$

Polynomial Arithmetic over GF(2)

$$\begin{array}{r}
 \begin{array}{r}
 x^7 \quad + x^5 + x^4 + x^3 \quad + x + 1 \\
 \times (x^3 \quad + x + 1) \\
 \hline
 x^7 \quad + x^5 + x^4 + x^3 \quad + x + 1 \\
 x^8 \quad + x^6 + x^5 + x^4 \quad + x^2 + x \\
 x^{10} \quad + x^8 + x^7 + x^6 \quad + x^4 + x^3 \\
 \hline
 x^{10} \quad \quad \quad + x^4 \quad + x^2 \quad + 1
 \end{array}
 \end{array}$$

(c) Multiplication

$$\begin{array}{r}
 \begin{array}{r}
 x^4 + 1 \\
 \hline
 x^3 + x + 1 \overline{) x^7 \quad + x^5 + x^4 + x^3 \quad + x + 1} \\
 \underline{x^7 \quad + x^5 + x^4} \\
 x^3 \quad + x + 1 \\
 \underline{x^3 \quad + x + 1} \\
 0
 \end{array}
 \end{array}$$

(d) Division

Perform following polynomial operations over GF(7)

- Let $f(x) = 8x^2 + 3x + 2$ and $g(x) = 2x + 1$ over the GF(7)
- Compute $f(x)+g(x)$, $f(x)-g(x)$, $f(x) * g(x)$ and $f(x)/g(x)$
- $f(x) + g(x) = 8x^2 + 3x + 2 + 2x + 1 = x^2 + 5x + 3$
- $f(x) - g(x) = 8x^2 + 3x + 2 - 2x - 1 = x^2 + x + 1$

Perform following polynomial operations over GF(7)

- Let $f(x) = 8x^2 + 3x + 2$ and $g(x) = 2x + 1$
- $f(x) * g(x) = (8x^2 + 3x + 2)(2x + 1)$
- $= 2x^3 + 6x^2 + 4x + 8x^2 + 3x + 2$
- $= 2x^3 + 2$

Perform following polynomial operations over GF(7)

2x+1	$\begin{array}{r} 8x^2 + 3x + 2 \\ 8x^2 + 4x \\ \hline \end{array}$	4x+3
	$\begin{array}{r} 6x + 2 \\ 6x + 3 \\ \hline \end{array}$	
	6	

GCD of polynomials over field

- Let $c(x)$ be gcd of two polynomials $a(x)$ and $b(x)$
- Then $c(x)$ has highest degree among all divisors of $a(x)$ and $b(x)$.
- $c(x)|a(x)$ and $c(x)|b(x)$
- $\gcd(a(x), b(x)) = \gcd(b(x), a(x) \bmod b(x))$
- So, we can use Euclidean algorithm to compute gcd of two polynomials.

GCD of polynomials over field

Euclidean Algorithm for Polynomials	
Calculate	Which satisfies
$r_1(x) = a(x) \bmod b(x)$	$a(x) = q_1(x)b(x) + r_1(x)$
$r_2(x) = b(x) \bmod r_1(x)$	$b(x) = q_2(x)r_1(x) + r_2(x)$
$r_3(x) = r_1(x) \bmod r_2(x)$	$r_1(x) = q_3(x)r_2(x) + r_3(x)$
• • •	• • •
$r_n(x) = r_{n-2}(x) \bmod r_{n-1}(x)$	$r_{n-2}(x) = q_n(x)r_{n-1}(x) + r_n(x)$
$r_{n+1}(x) = r_{n-1}(x) \bmod r_n(x) = 0$	$r_{n-1}(x) = q_{n+1}(x)r_n(x) + 0$ $d(x) = \gcd(a(x), b(x)) = r_n(x)$

GCD of polynomials over GF(2)

Find $\gcd[a(x), b(x)]$ for $a(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$ and $b(x) = x^4 + x^2 + x + 1$. First, we divide $a(x)$ by $b(x)$:

$$\begin{array}{r}
 x^2 + x \\
 \hline
 x^4 + x^2 + x + 1 \overline{) x^6 + x^5 + x^4 + x^3 + x^2 + x + 1} \\
 \underline{x^6 + x^4 + x^3 + x^2} \\
 x^5 + x + 1 \\
 \underline{x^5 + x^3 + x^2 + x} \\
 x^3 + x^2 + 1
 \end{array}$$

This yields $r_1(x) = x^3 + x^2 + 1$ and $q_1(x) = x^2 + x$.

Then, we divide $b(x)$ by $r_1(x)$.

$$\begin{array}{r}
 x + 1 \\
 \hline
 x^3 + x^2 + 1 \overline{) x^4 + x^2 + x + 1} \\
 \underline{x^4 + x^3 + x} \\
 x^3 + x^2 + 1 \\
 \underline{x^3 + x^2 + 1} \\
 0
 \end{array}$$

This yields $r_2(x) = 0$ and $q_2(x) = x + 1$.

Therefore, $\gcd[a(x), b(x)] = r_1(x) = x^3 + x^2 + 1$.

Problems with Galois Field

- In cryptography, the fields of form $GF(p^n)$ is extensively used.
- But $GF(p^n)$ is only field when $n=1$.
- For Example $GF(2^3)$ or $GF(8)$ is not a field.
- We can make it clear using the addition and multiplication table.
- The first operation addition modulo 8 satisfies all 5 properties.
- However, the second operation multiplication modulo 8 does not satisfy inverse property.

Problem with Galois Field

Table 4.2 Arithmetic Modulo 8

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

(a) Addition modulo 8

Problem with Galois Field

- All integers in the set Z_8 does not have multiplicative inverse modulo 8 as shown below

$\times$	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	0	2	4	6
3	0	3	6	1	4	7	2	5
4	0	4	0	4	0	4	0	4
5	0	5	2	7	4	1	6	3
6	0	6	4	2	0	6	4	2
7	0	7	6	5	4	3	2	1

(b) Multiplication modulo 8

Problem with Galois Field

- Moreover, the elements are not uniform under multiplication modulo 8.

$\times$	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	0	2	4	6
3	0	3	6	1	4	7	2	5
4	0	4	0	4	0	4	0	4
5	0	5	2	7	4	1	6	3
6	0	6	4	2	0	6	4	2
7	0	7	6	5	4	3	2	1

(b) Multiplication modulo 8

One solution

- Such unevenly mapping makes cryptography weaker.
- One thing we can do is we can select p as the largest prime number less than 2^n .
- But, this is inefficient because we cannot use the integers from p to $2^n - 1$.
- For example, if $n = 4$, the largest prime less than 16 is 13.
- This means that we cannot use integers 13, 14, and 15.

Another Solution

- As we know $\text{GF}(p^n)$ is not a field under normal integer modulo.
- In order to make $\text{GF}(p^n)$, we define two refinements to create $\text{GF}(p^n)$
- Create set of polynomials that $\text{GF}(p^n)$ can represent.
- $\text{GF}(p^n)$ can represent polynomials of max degree $n-1$
- So, n is number of bits used to represent the coefficients of polynomial.
- Multiplication is polynomial modulo operation

Extended Galois Field

- $GF(p^n)$ is an extended galois field define over set of polynomials of max degree $n-1$ of the form
- $a_{n-1}x^{n-1} + \dots + a_1x^1 + a_0x^0$
- Where p is a prime, n is positive integer.
- Set of coefficient is **$\{0, 1, \dots, p-1\}$**
- Addition and multiplication operation is **modulo p**
- Addition of polynomials is normal.
- Multiplication of polynomials is modulo some **irreducible polynomial** of degree n .

Irreducible Polynomial

- A polynomial $f(x)$ is called **irreducible** if and only if $f(x)$ cannot be expressed as a product of two polynomials both of degree lower than that of $f(x)$.
- $f(x) = x^3 + x + 1$ is irreducible.
- $f(x) = x^4 + 1$ is reducible, because $x^4 + 1 = (x + 1)(x^3 + x^2 + x + 1)$.
- By analogy to integers, an irreducible polynomial is also called a prime polynomial.

Irreducible Polynomial

Table 4.4 *List of irreducible polynomials*

<i>Degree</i>	<i>Irreducible Polynomials</i>
1	$(x + 1), (x)$
2	$(x^2 + x + 1)$
3	$(x^3 + x^2 + 1), (x^3 + x + 1)$
4	$(x^4 + x^3 + x^2 + x + 1), (x^4 + x^3 + 1), (x^4 + x + 1)$
5	$(x^5 + x^2 + 1), (x^5 + x^3 + x^2 + x + 1), (x^5 + x^4 + x^3 + x + 1),$ $(x^5 + x^4 + x^3 + x^2 + 1), (x^5 + x^4 + x^2 + x + 1)$

Creating set of polynomials

- We need to create set of polynomials for the field $GF(p^n)$
- To create set of polynomials, first define the form of polynomial
- $a_{n-1}x^{n-1} + \dots + a_1x^1 + a_0x^0$
- Now, substitute possible values of coefficients to the polynomial.
- Let us create a polynomial for a field where $p=2$ and $n=2$.

Extended GF(2^2)

- So max possible degree of polynomials is $n-1=1$
- So, polynomials are of the form $a_1x^1+a_0$
- The set of coefficient are $\{0,1\}$ since $p=2$.
- Substituting possible values of coefficient

a0	a1	polynomial
0	0	0
0	1	1
1	0	x
1	1	x+1

Extended $\text{GF}(2^2)$

- So, the set of polynomials is
- $\{0, 1, x, x+1\}$
- In binary $\{00, 01, 10, 11\}$
- In decimal $\{0, 1, 2, 3\}$

Extended GF(2²)

- We xor the coefficients for addition.
- We can perform addition modulo 2 for coefficients.
- All the elements of the set have additive inverse.
- All the elements are uniform.

Polynomial table

+	0	1	x	1+x
0	0	1	x	1+x
1	1	0	1+x	x
x	x	1+x	0	1
1+x	1+x	x	1	0

Additive Table in Binary

+	00	01	10	11
00	00	01	10	11
01	01	00	11	10
10	10	11	00	01
11	11	10	01	00

Extended GF(2^2)

- For multiplicative table, remember that multiplication may result in the polynomials of degree more than $(n-1)$
- In such case, we need to perform polynomial modulo with some **irreducible polynomial** of degree n

Extended GF(2²)

- For field GF(2²), we use irreducible polynomial x^2+x+1
- Whenever degree of polynomial after multiplication is more than 1 (n-1), we perform modulo polynomial x^2+x+1

Polynomial table

x	0	1	x	1+x
0	0	0	0	0
1	0	1	x	1+x
x	0	x	1+x	1
1+x	0	1+x	1	x

Multiplicative Table in Binary

x	00	01	10	11
00	00	00	00	00
01	00	01	10	11
10	00	10	11	01
11	00	11	01	10

Extended $\text{GF}(2^2)$

- This shows that $\langle \{0,1,x,1+x\}, +, \cdot \rangle$ is a field under modular arithmetic with irreducible polynomial x^2+x+1
- So $\text{GF}(2^2)$ supports polynomial arithmetic modulo x^2+x+1

Extended GF(2^3)

		000	001	010	011	100	101	110	111
	+	0	1	2	3	4	5	6	7
000	0	0	1	2	3	4	5	6	7
001	1	1	0	3	2	5	4	7	6
010	2	2	3	0	1	6	7	4	5
011	3	3	2	1	0	7	6	5	4
100	4	4	5	6	7	0	1	2	3
101	5	5	4	7	6	1	0	3	2
110	6	6	7	4	5	2	3	0	1
111	7	7	6	5	4	3	2	1	0

(a) Addition

Extended GF(2^3)

		000	001	010	011	100	101	110	111
	$\times$	0	1	2	3	4	5	6	7
000	0	0	0	0	0	0	0	0	0
001	1	0	1	2	3	4	5	6	7
010	2	0	2	4	6	3	1	7	5
011	3	0	3	6	5	7	4	1	2
100	4	0	4	3	7	6	2	5	1
101	5	0	5	1	4	2	7	3	6
110	6	0	6	7	1	5	3	2	4
111	7	0	7	5	2	1	6	4	3

(b) Multiplication

Extended Galois Field

- Add $(x^5 + x^2 + x)$ and $(x^3 + x^2 + 1)$ in $GF(2^8)$.

$$\begin{array}{rcl} 0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0 & \oplus & \\ 0x^7 + 0x^6 + 0x^5 + 0x^4 + 1x^3 + 1x^2 + 0x^1 + 1x^0 & & \\ \hline 0x^7 + 0x^6 + 1x^5 + 0x^4 + 1x^3 + 0x^2 + 1x^1 + 1x^0 & \rightarrow & x^5 + x^3 + x + 1 \end{array}$$

Extended Galois Field

- Multiply polynomials $(x^5 + x^2 + x)$ and $(x^7 + x^4 + x^3 + x^2 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$.

$$P_1 \otimes P_2 = x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x)$$

$$P_1 \otimes P_2 = x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2$$

$$P_1 \otimes P_2 = (x^{12} + x^7 + x^2) \bmod (x^8 + x^4 + x^3 + x + 1) = x^5 + x^3 + x^2 + x + 1$$

Operation in binary form

- For polynomials of form $GF(2^n)$, we can easily perform operation in binary form.
- For this, convert the polynomials in binary form.
- The addition and subtraction operation is XOR operation.
- Add $(x^5 + x^2 + x)$ and $(x^3 + x^2 + 1)$ in $GF(2^8)$.
- 00100110
- 00001101
- 00101011
- This is the polynomial x^5+x^3+x+1

Multiplication in binary form

- Convert the highest degree polynomials $f(x)$ in binary form $b_{n-1}b_{n-2}\dots b_1b_0$
- Multiply it with polynomials of by 2, 4, 8 upto $n-1$
- The multiplication simply

$$x \times f(x) = \begin{cases} (b_6b_5b_4b_3b_2b_1b_00) & \text{if } b_7 = 0 \\ (b_6b_5b_4b_3b_2b_1b_00) \oplus (00011011) & \text{if } b_7 = 1 \end{cases}$$

- Now, convert $g(x)$ in binary form.
- Express $g(x)$ in xor form.
- Distribute $f(x)$ to xor of $g(x)$.

Extended Galois Field

- Multiplication in binary form

<i>Powers</i>	<i>Shift-Left Operation</i>	<i>Exclusive-Or</i>
$x^0 \otimes P_2$		10011110
$x^1 \otimes P_2$	00111100	$(00111100) \oplus (00011010) = \underline{\underline{00100111}}$
$x^2 \otimes P_2$	01001110	<u>01001110</u>
$x^3 \otimes P_2$	10011100	10011100
$x^4 \otimes P_2$	00111000	$(00111000) \oplus (00011010) = 00100011$
$x^5 \otimes P_2$	01000110	<u>01000110</u>
$P_1 \otimes P_2 = (00100111) \oplus (01001110) \oplus (01000110) = 00101111$		

Multiplication in binary form

- Multiply polynomials $(x^5 + x^2 + x)$ and $(x^7 + x^4 + x^3 + x^2 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$.

Powers	One poly	Multi	irreduc	result
00000001	10011110	10011110		
00000010	10011110	00111100	00011011	00100111
00000100	00100111	01001110		
00001000	01001110	10011100		
00010000	10011100	00111000	00011011	00100011
00100000	00100011	01000110		
01000000	01000110	10001100		
10000000	10001100	00011000	00011011	00000011

Multiplication in binary form

- Multiply polynomials $(x^5 + x^2 + x)$ and $(x^7 + x^4 + x^3 + x^2 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$.
- 00100110×10011110
- $= 10011110 (00100000 + 00000100 + 00000010)$
- $= 10011110 \times 00100000 + 10011110 \times 00000100 + 10011110 \times 00000010$
- $= 01000110 + 01001110 + 00100111$
- $= 00101111$
- $= x^5 + x^3 + x^2 + x + 1$

01000110
01001110
00100111
00101111

Multiplication in binary form

In an earlier example, we showed that for $f(x) = x^6 + x^4 + x^2 + x + 1$, $g(x) = x^7 + x + 1$, and $m(x) = x^8 + x^4 + x^3 + x + 1$, we have $f(x) \times g(x) \bmod m(x) = x^7 + x^6 + 1$. Redoing this in binary arithmetic, we need to compute $(01010111) \times (10000011)$. First, we determine the results of multiplication by powers of x :

$$(01010111) \times (00000010) = (10101110)$$

$$(01010111) \times (00000100) = (01011100) \oplus (00011011) = (01000111)$$

$$(01010111) \times (00001000) = (10001110)$$

$$(01010111) \times (00010000) = (00011100) \oplus (00011011) = (00000111)$$

$$(01010111) \times (00100000) = (00001110)$$

$$(01010111) \times (01000000) = (00011100)$$

$$(01010111) \times (10000000) = (00111000)$$

So,

$$\begin{aligned}(01010111) \times (10000011) &= (01010111) \times [(00000001) \oplus (00000010) \oplus (10000000)] \\ &= (01010111) \oplus (10101110) \oplus (00111000) = (11000001)\end{aligned}$$

which is equivalent to $x^7 + x^6 + 1$.

Multiplicative inverse

- In $GF(2^4)$, find the inverse of x^2+1 modulo x^4+x+1
- This shows that $\gcd(x^4+x+1, x^2+1)$ is 1.
- Multiplicative inverse is x^3+x+1

$q(x)$	$a(x)$	$b(x)$	$r(x)$	$t1(x)$	$t2(x)$	$t(x)=t1(x)-t2(x)*q(x)$
X^2+1	x^4+x+1	X^2+1	x	0	1	X^2+1
x	X^2+1	x	1	1	X^2+1	x^3+x+1
x	x	1	0	X^2+1	x^3+x+1	0
	1	0		x^3+x+1	0	

00011000

00011011

00000011

Construct field with generator

- We can easily define the elements of $GF(2^n)$ field using a generator.
- Every $GF(2^n)$ field has a generator g .
- If the irreducible polynomial is $f(x)$, then $f(g)=0$
- The elements of the field can be generated as
- $\{0, g, g, g^2, \dots, g^N\}$, where $N = 2^n - 2$

Construct field with generator

- We have seen that elements of a field $GF(2^2)$ are
- $\{0, 1, x, x+1\}$
- Let us generate these elements using generator g .
- The irreducible used is $f(x)=x^2+x+1$
- $N=2^2-2=4$
- Now, $f(g)=0$ or $g^2+g+1=0$ or $g^2=-g-1$
- Since subtraction is same to addition modulo 2
- $g^2=g+1$

Construct field with generator

- $g^3 = gg^2 = (g+1)g = g^2 + g = g+1+g = 1$
- $g^4 = gg^3 = g \cdot 1 = g$
- $g^5 = gg^4 = g(g) = g^2 = g+1$
- We can see that it generate all the elements of the field except 0.
- So, elements are
- $\{0, 1, g, g+1\}$

Construct field with generator

- Let us generate elements of another field $GF(2^3)$
- We know the elements are $\{0, 1, x, x+1, x^2, x^2+1, x^2+x, x^2+x+1\}$
- Let us use the irreducible polynomial $f(x)=x^3+x+1$
- Let the generator element be g
- $f(g)=g^3+x+1=0$
- Or $g^3=-x-1$
- since subtraction and addition modulo 2 is same
- Or $g^3=x+1$

Construct field with generator

- Now we will calculate next power
- $g^4 = g(g^3) = g(g+1) = g^2 + g$
- $g^5 = g(g^4) = g(g^2 + g) = g^3 + g^2 = g + 1 + g^2$
- $g^6 = g(g^5) = g(g^2 + g + 1) = g^3 + g^2 + g = g + 1 + g^2 + g = g^2 + 1$
- $g^7 = g(g^6) = g(g^2 + 1) = g^3 + 1 = g + 1 + 1 = 1 = g^0$
- So, elements of the field $GF(2^3)$ are
- $\{0, g^0, g^1, g^2, g^3, g^4, g^5, g^6\}$
- $\{0, 1, g, g^2, g+1, g^2+g, g^2+g+1, g^2+1\}$